Slowing the None-Critical Path to Improve Carry Look-Ahead Adder Power Dissipation

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***Abstract*—In this paper we use a 4-bit carry look-ahead adder to highlight the contribution by false-starts (glitches) to overall dynamic power dissipation. These false-starts occur in the generation of the sum outputs and are due to delays in generating and propagating the carry signals. We employ sub-threshold transistor operation in the none critical path and reduce power dissipation by 40%. Post layout simulations in a 90 nm technology node have been performed and for 1.8 % increase in transistor count we improve power dissipation by 40 %. It is easy to overlook the amount of power dissipation due to false-starts because input and output latches that are typically employed in most synchronous digital systems/circuits mask these glitches. By sampling the signals after they settle at the desired steady state values we can avoid sending the glitches through latches to the sum outputs. This does not however eliminate the power dissipated by the circuit.**

Keywords—power dissipation; false-start; post layout; Critical path; subthreshold; none critical path, Carry Look-Ahead Adder

I. INTRODUCTION

Adders are basic components in the design of many computing elements that include general purpose Central Processing Units (CPU), Digital Signal Processors (DSP), Graphic Processing Units (GPU), Floating Point Units (FPU), just to name a few. Most of the computing units listed above employ wide adders [1] which perform operations in circuits and/or systems that require high speed and low-power. Speed and power are conflicting performance metrics in that the desirable act of increasing speed/frequency of operations for a design often results in increasing power, on the other hand lowering power dissipation would result in a decrease in speed. There is always some trade-off when designing for speed and power. Improving power dissipation for adder circuits has been the subject of study by many researchers [2][3][4][5], yet others have extensively worked on improving adder circuits' speed [6][7]. Some work has sought to address both speed and power for adder circuits.

Improved adder performance has been sought at all levels of digital circuits and systems design. With the advent of low-voltage battery operated digital circuits minimizing power dissipation has become critical in adder design. This in an effort to deliver circuits and systems that will have prolonged operations on battery power.

None of the papers that focus on Carry Look-Ahead Adder circuits and architectures address the fact that the carry signals of the 4-bit building block for wide adders, when transitioning have delays and depending on the bit position producing the carry; the delay leads to incorrect sums being initially produced. The carry bit's propagation delay influences the duration of the incorrect sum outputs and they can appear as brief as spikes or they can maintain steady state values as pulses with considerable pulse widths. We explore the amount of power dissipated by a 4-bit carry look-ahead adder particularly with instances that lead to the false sum outputs. The false sum outputs occur for some input combinations and not for all.

This paper is organized as follows: Section II presents the general carry look-ahead adder circuit schematics and associated equations, Section III discusses the proposed none critical path subthreshold delay balancing circuitry and presents the power savings earned with Section IV providing some concluding remarks.

II. CARRY LOOK-AHEAD ADDER CIRCUITS AND ASSOCIATED EQUATIONS

A. Binary Addition:-Full Adder Description

Half and full adder circuits form the basic building blocks of wide adders. From the basic addition circuitry in a bid to improve operation speed and lower power many variations of the adder circuits have been developed. Binary addition of two 1-bit numbers forms the basis in developing the Boolean expressions for the sum and carry outputs of a half adder. Addition of two n-bit binary numbers reveals that each bit position after bit 0 might add a carry bit from the bit position preceding it, resulting in the addition of three 1-bit numbers instead. The full adder accommodates the possibility of the occurrence of a carry bit being generated or propagated from the preceding bit positions. We concentrate on the full adder to produce the Boolean expressions for the sum and carry signals. This allows us to provide the needed background in analyzing the carry look-ahead adder considered to be a fast adder among the various adder design choices. Table 1 presents a truth table that depicts the addition of three 1-bit numbers.

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TABLE I. FULL ADDER TRUTH TABLE

Minterm	INPUTS			OUTPUTS	
	A	B	C	Sum	Carry
0	0	0	0	0	0
1	0	0	1	1	0
2	0	1	0	1	0
3	0	1	1	0	1
4	1	0	0	1	0
5	1	0	1	0	1
6	1	1	0	0	1
7	1	1	1	1	1

The Boolean expressions for the sum and carry outputs are:

$$Sum = \bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC \quad (1)$$

$$Carry = \bar{A}BC + A\bar{B}C + AB\bar{C} + ABC \quad (2)$$

Equation 1 simplifies to $Sum = A \oplus B \oplus C$, thus the sum is widely implemented using XOR gates. The fast implementation of XOR gates in CMOS technologies employs the pass transistor logic family. Equation 2 can be simplified to be: $Carry = (A \oplus B) \bullet C + AB$.

The circuits constructed based on the simplified versions of equations 1 and 2 can be cascaded to build an n-bit adder as depicted in the block diagram shown in Figure 1.

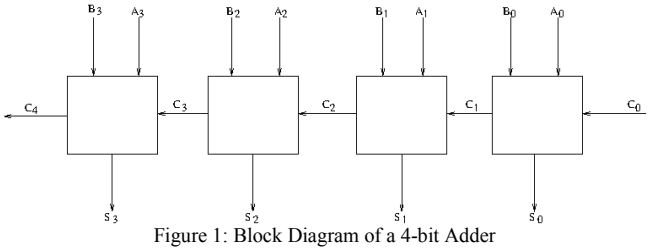


Figure 1: Block Diagram of a 4-bit Adder

The block diagram of Figure 1 shows that with the exception of bit position 0 where inputs A_0, B_0 and C_0 arrive at the same time (assuming latches are employed to admit the inputs) the subsequent bit positions have as part of their inputs a carry bit generated or propagated from the addition of the preceding bit position. The carry C_1 needed to produce $S_1 = A_1+B_1+C_1$ is produced from the addition of A_0, B_0, and C_0 while C_2 needed to produce $S_2 = A_2 + B_2 + C_2$ is obtained only after summing A_1, B_1 and C_1. This implies that there is potential for a longer delay in producing C_3 the carry signal required in producing sum S_3. Designers of carry look-ahead adders mitigate the carry bit propagation delay by observing from Table 1 that the four cases that produce a carry signal have two unique conditions, the first occurs whenever input *A* OR input *B* is at logic 1 AND input *C* is at logic 1. This first condition of the two cases is represented by the expression:

$$p = A \oplus B \quad (3)$$

The second condition occurs when $A = B = 1$ and we note that this condition generates a carry irrespective of the logic level of input *C*. It is thus termed *generate* and is expressed as:

$$g = AB \quad (4)$$

Revisiting Figure 1, we can write a general equation that determines the expressions for the sum and carry outputs for any bit position of an n-bit adder. The expressions for the sum and carry signals for the ith bit position of an n-bit adder are:

$$S_i = A_i \oplus B_i \oplus C_i \quad (5)$$

$$C_i = (A_{i-1} \oplus B_{i-1}) \bullet C_{i-1} + A_{i-1}B_{i-1} \quad (6)$$

The expression for the carry input of the ith bit position is dependent on the inputs of the preceding bit position as described by equation 6.

B. 4-Bit Carry Look-Ahead Adder Carry Computations

Using the general expression we can write the sum and carry expressions for each of the four outputs of a 4-bit adder.

$$S_0 = A_0 \oplus B_0 \oplus C_0 \quad (7)$$

C_0 for bit position 0 is ideally 0 except for cases when operations being performed involve computations in 2's complement.

$$S_1 = A_1 \oplus B_1 \oplus C_1 \quad (8)$$

$$C_1 = (A_0 \oplus B_0) \bullet C_0 + A_0B_0$$
$$C_1 = p_0C_0 + g_0 \quad (9)$$

We have substituted $(A_0 \oplus B_0)$ by p_0 and A_0B_0 by g_0.
Figure 2 shows a three level logic circuit that would generate the carry required at bit position 1.

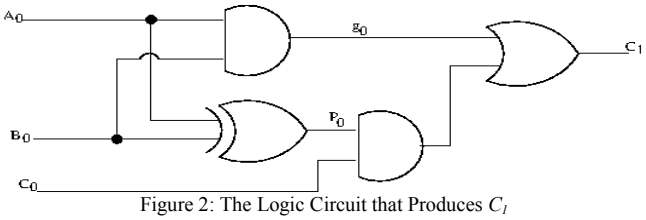


Figure 2: The Logic Circuit that Produces C_1

The circuit of Figure 2 shows that the desired logic level of C_1 is determined after propagating through a path of a three level logic circuit. This affects S_1 in that when new inputs are admitted A_1 and B_1 are available long before the new C_1 can be produced and propagated to bit position 1. A sum output, $A_1 \oplus B_1 \oplus C_1$, produces a new output, however C_1 is an old value from the previous computation. Until the newly generated or propagated C_1 arrives the sum output is incorrect, the resulting switching activity in producing the false sum and then switching to the correct sum upon arrival of the current value of C_1 dissipates power. The dissipated power for a single bit is negligible but for wide adders with many false starts per cycle the dissipated power becomes significant. The impact of the power dissipated due to these glitches will likely be felt more in low-voltage battery operated electronic devices.

The sum of the third bit position has its expression depicted in equation 10 and the carry input to this bit position

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is C_1 determined in the preceding bit position. C_1 has already imposed a delay on S_1 and would impose an even longer delay if used in the generation of S_2. The carry look-ahead adder minimizes this delay by using $(A_0 \oplus B_0) \cdot C_0 + A_0 B_0$ in place of C_1. The delay is reduced since p_0, g_0 and C_0 are produced at the same time with $p_2 = A_2 \oplus B_2$. Ideally the delay incurred is identical to that of C_1 because the circuits that produce C_1 and C_2 have the same depth i.e. they both have three levels of logic. We note here that the logic circuit that computes C_2 has two gates with a fan-in of 3 while those that compute C_1 have a fan-in of 2 and this can lead to delay differences between these two signals if not addressed with proper device sizing at the layout level.

$$S_2 = A_2 \oplus B_2 \oplus C_2 \tag{10}$$

$$C_2 = (A_1 \oplus B_1) \bullet C_1 + A_1 B_1$$
$$C_2 = p_1 C_1 + g_1 \tag{11}$$

The sum S_3 is computed in a similar manner as S_0, S_1, and S_2 and C_3 has the same logic depth as C_1 and C_2 but with some of the gates having a fan-in of 4. Equations 12 and 13 describe the sum and carry signals of this bit position.

$$S_3 = A_3 \oplus B_3 \oplus C_3 \tag{12}$$

$$C_3 = (A_2 \oplus B_2) \bullet C_2 + A_2 B_2$$
$$C_3 = p_2 C_2 + g_2 \tag{13}$$

The focus is on a 4-bit carry look-ahead adder's false sums and an examination of their contribution to over-all dynamic power dissipation, we thus mention in passing that the fan-in of 4 of C_3 dictates that C_4 of a carry look-ahead adder cannot be generated the same way as C_1, C_2 and C_3 because it will have a restrictive fan-in of 5. Wide carry look-ahead adders are designed by using 4-bit carry look-ahead adders as building blocks. The 5th carry signal (C_4) counting form C_0 is computed from a group of 4 carry signals.

III. Power Dissipation Due to False Sums

An example of the problem of false sums due to the delay in generating or propagating the carry signal is demonstrated by adding 15_{10} and 1_{10} in their binary equivalents. Table 2 shows that each bit position produces a carry signal that is added to the succeeding bit.

TABLE II. Worst Case Scenario Addition

	Bit_4	Bit_3	Bit_2	Bit_1	Bit_0
Carries	1	1	1	1	0
Input A		1	1	1	1
Input B		+0	+0	+0	+1
Sums		0	0	0	0

Layouts of two 4-bit carry look-ahead adders using Synopsys' suite of CAD Tools were completed in a 90 nm technology node. The layouts differ in that the reference layout does not have subthreshold circuitry while the improved layout has delay balancing circuitry in its non-critical path circuitry. Latches have been employed at the input port to admit data and at the output port to sample the outputs at the desired times. The Sum signals are sampled before and after the latches and compared for glitches.

Delays to eliminate false sum outputs are achieved by taking advantage of the fact that NMOS transistors are poor at transmitting logic 1s while PMOS transistors are poor at transmitting logic 0s. We thus create a separate power rail for the adder circuits in the none critical path. We introduce a long enough delay to match the delay experienced in generating the corresponding carry for each bit position. The additional circuit only has six transistors for each sum in the sum blocks except for the sum_0 bit and we show just two in Figure 3.

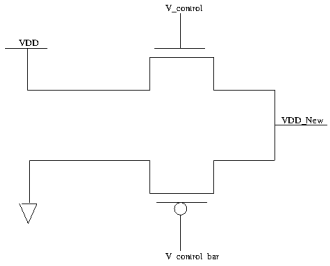


Figure 3: Simple effective Circuit for Delaying None Critical Path

The control signal and its inverse are designed to provide the appropriate operating conditions to generate a new V_{DD} rail whose value ranges from $|V_{tp}| \leq V_{DD_{NEW}} \leq V_{DD} - V_{tn}$. Where V_{tp} is the threshold voltage of the PMOS, V_{tn}, the NMOS threshold voltage, V_{DD_New} the desired power rail and V_{DD} the power rail for the rest of the design. If the control signal is at logic 0 the PMOS device drives the output and the minimum voltage it can reach is V_{tp}. Collapsing the V_{DD_New} rail to this value allows the delay circuit to operate in the subthreshold voltages. If the control input is at logic one the output is driven to $V_{DD} - V_{tn}$ and the circuit operates at above threshold voltages. We can also run the circuit at any desirable intermediate voltages. Post Layout simulations corresponding to the reference layout appear in Figure 4.

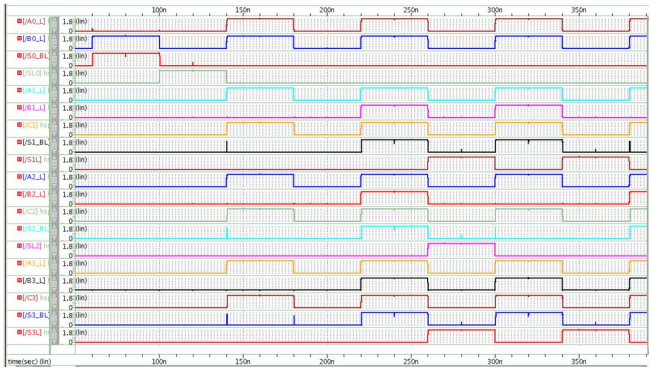


Figure 4: The False Sums for a Select Inputs

Figure 5 shows the pulse widths of the false sums. The sum of bit position 0 (S_0) does not produce a false-start because the inputs including the carry bit are all available without any significant delays. S_1, S_2 and S_3 all switch in response to some of the changing inputs, but until the carry signals corresponding to the current inputs are generated the sum

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outputs are incorrect and they dissipate dynamic power. The traces shown in Figure 5 are viewed before the latches i.e. before the sum outputs are clocked. We note that with regard to power dissipation, it is not the duration of the pulses that matter most for these glitches, but their amplitudes and the frequency of their occurrence.

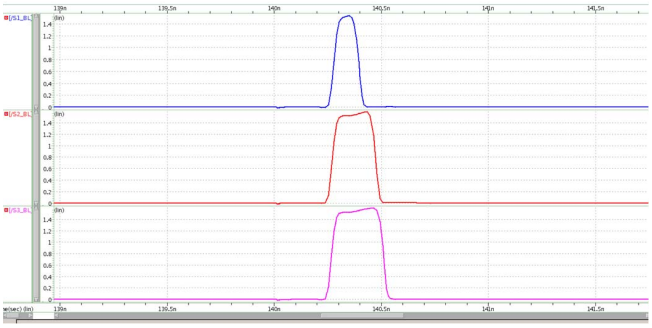


Figure 5: The Pulse Widths of the False Sums

In Table III we list a few representative pairs of 4-bit numbers that were added during the post layout simulations for both circuits.

TABLE III. REPRESENTATIVE PAIRS OF THE ADDED NUMBERS

A				B				Decimal
0	0	0	0	0	0	0	0	0 + 0
0	0	0	0	0	0	0	1	0 + 1
1	1	1	1	0	0	0	1	15 + 1
1	1	1	1	1	1	1	1	15 + 15
1	1	1	1	1	0	1	1	15 + 11

Figure 6 shows the simulations with delay insertion and the sums sampled both before and after the latches have no glitches. We have also included in Figure 6 the new power rail generated by the subthreshold circuitry. The new V_{DD} has an amplitude of 1.4 Volts in this design, but can be adjusted to any desired amplitude through the V_control and V_control_bar inputs of Figure 3.

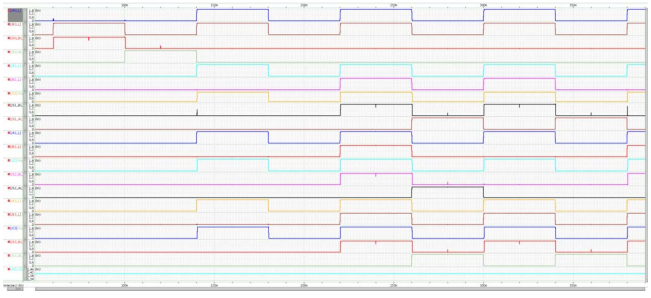


Figure 6: None Critical Path Slowed to Eliminate Glitches

Figure 7 is intended to be compared to Figure 5 to show how delaying the none critical path has minimized false-starts. Only S_1 shows a glitch whose amplitude is 50 % of VDD while S0, S2 and S3 suffer no false- start.

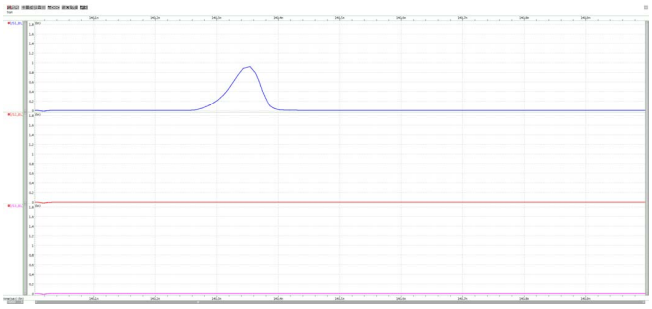


Figure 7: Sum False-Starts No Longer as Prominent as in Figure 5

The average current drawn from the power rail during the addition of eight different pairs of 4-bit numbers is 0.209 mA for the reference circuit while the proposed circuit that uses delay insertion in the non-critical path draws an average of 0.1254 mA.

IV. CONCLUDING REMARKS

A simple circuit has been designed to insert delays in the none critical path of a 4-bit carry look-ahead adder and it can be controlled to operate at voltages above threshold or in the subthreshold region to attain the desired delay. False sums that are normally masked by latches are eliminated resulting in 40 % power savings compared to the circuit without delay insertion. The penalty is an increase of the transistor count by 18, compared to 997 transistors of the 4-bit adder layout without delay insertion.

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